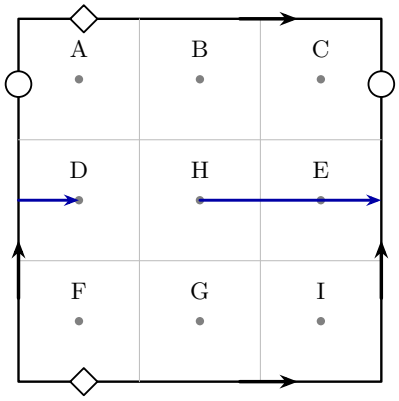
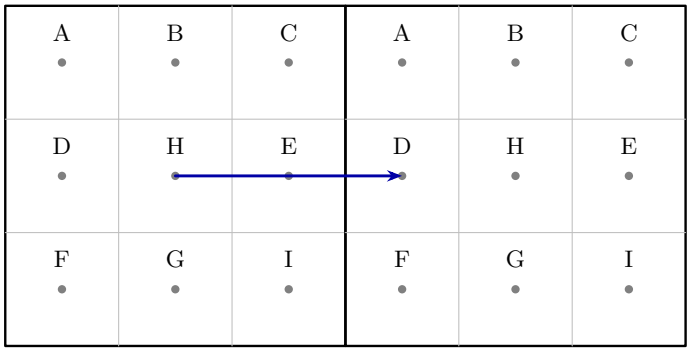
One arrow moves to the center of the next small square. Right meets left at the same height; top meets bottom at the same place across. Keep going the same way. Repeated copies continue beyond the page.

portal map



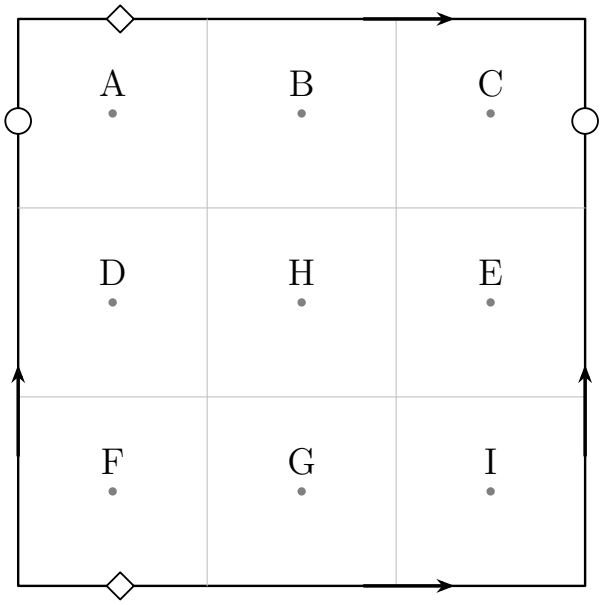
two steps right

repeated map



original copy next copy

Problem 1: Start at H for each trip. Predict where it ends, then try it with your pawn.



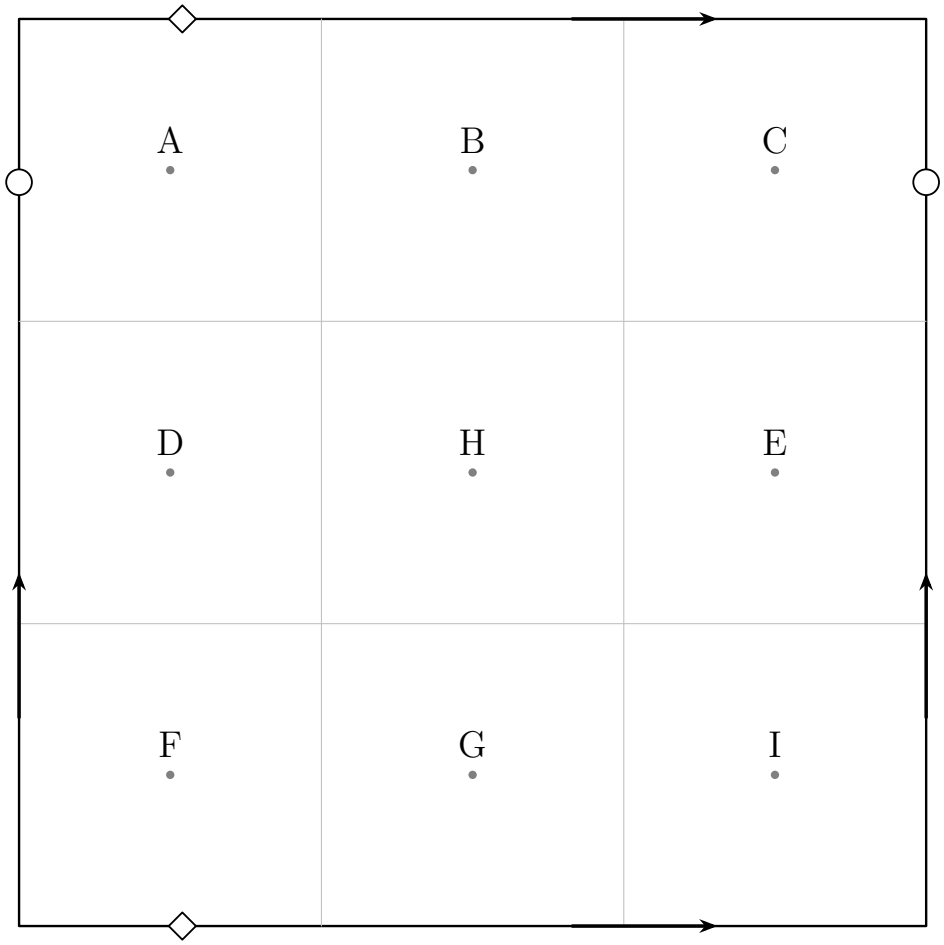
→ → ↓ _____

↑ ↑ _____

← ↓ _____

↑ ↑ → → _____

Problem 2: Find as many different trips from H back to H as you can, using exactly three steps, then exactly six steps.



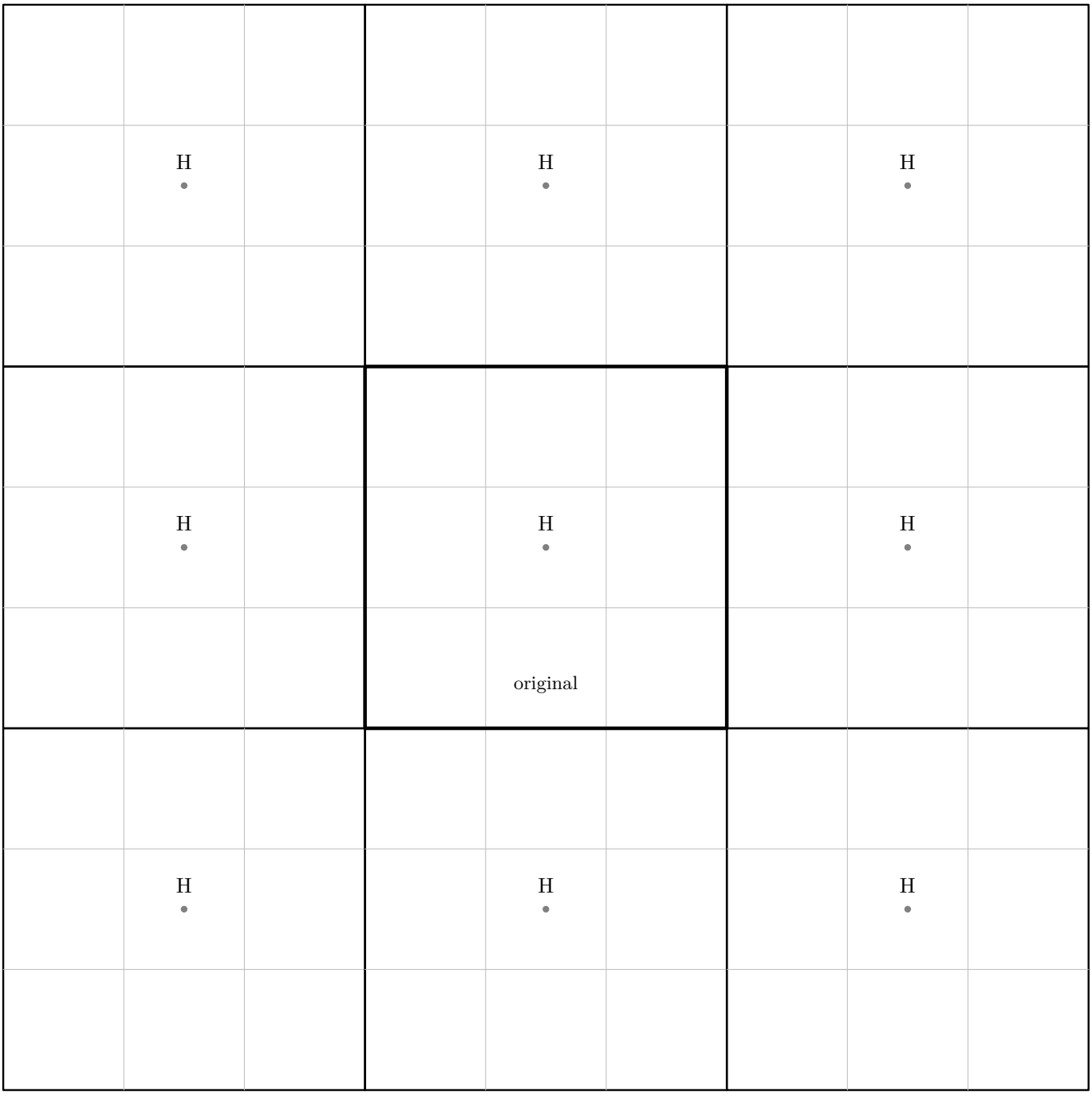
Problem 3: Replay each trip on these copies, starting at the original H. Mark the copy where each trip finishes.

$\rightarrow \rightarrow \rightarrow$

$\rightarrow \rightarrow \rightarrow \uparrow \uparrow \uparrow$

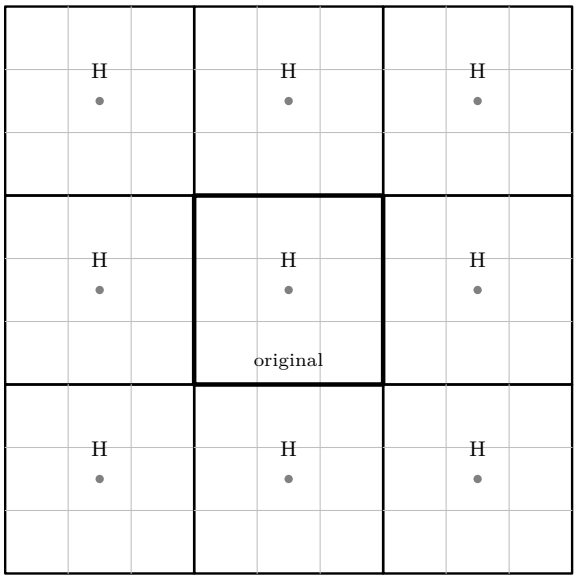
$\uparrow \uparrow \uparrow$

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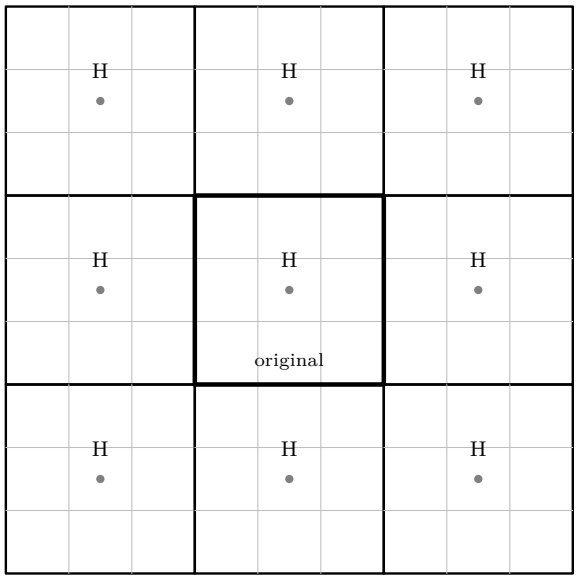


Problem 4: Compare these three trips from H. Which of them finish at the same place on the portal map? Which finish in the same copy on the repeated map?

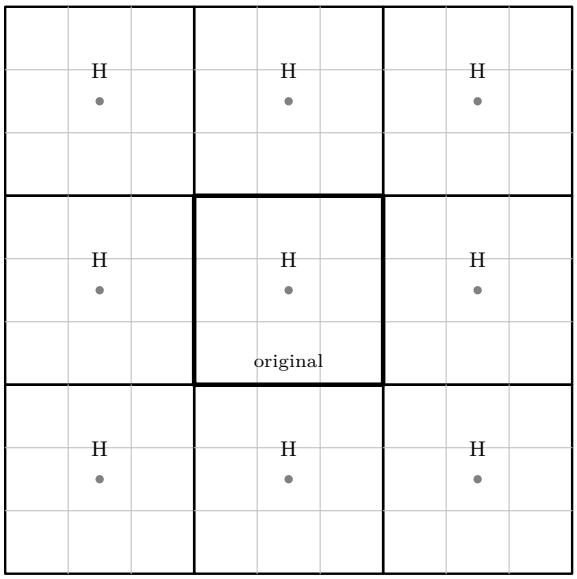
$\rightarrow \rightarrow \rightarrow \uparrow \uparrow \uparrow$



$\rightarrow \uparrow \rightarrow \uparrow \rightarrow \uparrow$



$\rightarrow \rightarrow \rightarrow \downarrow \downarrow \downarrow$



Problem 5: Make a trip that visits another copy and finishes at the original H. Can you make one that visits both a copy above and a copy to the right?

